

Denoised Gradient Estimation: Training Deep Neural Networks Without Backpropagation via Block Perturbations and Temporal Filtering

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Abstract

Gradient-based optimization requires analytical derivatives, limiting its applicability to differentiable architectures. We introduce **Denoised Gradient Estimation** (DGE), a zeroth-order algorithm that trains neural networks using only forward-pass evaluations. DGE combines three mechanisms: (1) *block-wise perturbations* that reduce gradient variance by a factor of D/K , where K is the block size; (2) *temporal exponential moving average* filtering that improves signal-to-noise ratio as $\mathcal{O}(\sqrt{T})$; and (3) a novel *Dual Sign-EMA* (DS-EMA) consistency mask inspired by MACD crossover detection, achieving 90% memory reduction over sliding-window approaches. On full MNIST (60K samples, 109K parameters), DGE achieves **94.36% $\pm$ 0.23%** test accuracy without backpropagation—reaching 96.3% of Adam’s performance. In non-differentiable settings where backpropagation fails entirely, DGE achieves 73–82% accuracy versus $\sim$ 9% for Adam on fully quantized (INT4/INT8) and sign-activation networks, without straight-through estimators. We provide theoretical analysis establishing the estimator’s unbiasedness and characterize the variance reduction and SNR improvement bounds.

1 Introduction

Backpropagation is the foundation of modern deep learning [Rumelhart et al., 1986, LeCun et al., 2015], but its reliance on analytical gradients limits applicability in several important regimes: (i) black-box systems where the forward pass is an opaque oracle, (ii) non-differentiable hardware such as neuromorphic or analog processors [Schuman et al., 2022, Davies et al., 2018], (iii) discrete architectures including binary [Yin et al., 2020] and ternary networks, and (iv) memory-constrained settings where storing activation graphs for backward passes is infeasible.

Zeroth-order methods estimate gradients using only function evaluations, offering a universal alternative. However, standard approaches face a fundamental scalability barrier. Simultaneous Perturbation Stochastic Approximation (SPSA) [Spall, 1992] perturbs all D parameters simultaneously, yielding an estimator whose variance scales linearly with dimensionality. For modern networks with $D > 10^5$ parameters, this variance overwhelms the gradient signal, causing the optimizer to diverge [Ghadimi et al., 2013].

Recent work has revisited zeroth-order optimization for fine-tuning large language models. MeZO [Malladi et al., 2023] achieves memory-efficient fine-tuning by regenerating perturbation vectors from a seed, but operates in the favorable regime where pre-trained models are near a local optimum—a setting fundamentally different from training from scratch.

Our approach. We propose Denoised Gradient Estimation (DGE), which resolves the dimensionality curse through three complementary mechanisms:

1. **Block perturbations:** Partition the D -dimensional parameter space into $K \ll D$ blocks per step, reducing gradient variance by a factor of D/K ($781\times$ for $D=100,000$, $K=128$).

2. **Temporal denoising:** Inject sparse gradient estimates into an Adam-style EMA filter, exploiting the fact that true gradient signals accumulate constructively while independent noise cancels destructively over time, yielding SNR improvement as $\mathcal{O}(\sqrt{T})$.
3. **Dual Sign-EMA (DS-EMA):** A MACD-inspired consistency gate that replaces sliding-window sign tracking with two exponential moving averages (fast $\alpha=0.3$, slow $\alpha=0.05$), achieving 90% memory reduction while improving robustness to non-stationary noise.

Contributions.

- A block-perturbation strategy with theoretical variance bound $\text{Var}(g_i) \leq (K/D)\|\nabla f\|^2$, enabling zeroth-order training of 100K+ parameter networks.
- DS-EMA: a novel temporal consistency mechanism that achieves superior noise suppression with constant memory.
- Empirical validation on MNIST (94.36% without gradients), INT8 quantization (82.20% vs. 8.40% for Adam), and sign-activation networks (73.20% vs. 61.20%).
- A systematic analysis of block size selection showing $\mathcal{O}(\sqrt{D})$ scaling is empirically optimal for neural network architectures.

2 Related Work

Finite differences. The classical approach computes exact gradients via central differences: $g_i = (f(x + \delta e_i) - f(x - \delta e_i))/(2\delta)$. This requires $\mathcal{O}(D)$ evaluations per step, making it computationally infeasible for neural networks with millions of parameters [Nocedal and Wright, 2006, Bertsekas, 2016].

SPSA and variants. SPSA [Spall, 1992] reduces evaluations to $\mathcal{O}(1)$ by perturbing all dimensions simultaneously with a shared Rademacher vector. The estimator is unbiased but its variance scales as $\mathcal{O}(D)$, rendering it ineffective for high-dimensional problems [Ghadimi et al., 2013]. Variants such as weighted SPSA and adaptive SPSA attempt to mitigate variance but retain the fundamental $\mathcal{O}(D)$ scaling.

Evolution strategies. OpenAI’s Evolution Strategies (ES) [Salimans et al., 2017] and Natural Evolution Strategies (NES) [Wierstra et al., 2014] use population-based search with Gaussian perturbations. While effective for reinforcement learning, they require $\mathcal{O}(\lambda)$ evaluations per generation for λ offspring, and their applicability to supervised learning with large datasets is limited by evaluation cost.

Forward gradients and memory-efficient fine-tuning. Forward-mode differentiation computes directional derivatives using dual numbers or perturbation vectors. MeZO [Malladi et al., 2023] exploits this for memory-efficient LLM fine-tuning, achieving competitive results with $\mathcal{O}(1)$ memory. However, MeZO is designed for the favorable regime of fine-tuning near-optimal models; it fails when training from scratch at scale, as we demonstrate experimentally.

Non-differentiable training. Training networks with discrete weights or activations typically requires straight-through estimators (STE) [Bengio et al., 2013], post-training quantization [Banner et al., 2019], or quantization-aware training with surrogate gradients. DGE circumvents these heuristics entirely by operating as a black-box optimizer.

3 Method

We consider the problem of minimizing $f : \mathbb{R}^D \rightarrow \mathbb{R}$ using only function evaluations, without access to ∇f .

3.1 Block Perturbations

At step t , DGE partitions the D -dimensional parameter space into K non-overlapping blocks via a random permutation π_t . For each block $B_k \subset \{1, \dots, D\}$, DGE generates a Rademacher vector $s_t^{(k)} \in \{-1, +1\}^{|B_k|}$ and evaluates the symmetric finite difference:

$$g_{t,i} = \frac{f(x_t + \delta \cdot s_t^{(k)} \odot m_k) - f(x_t - \delta \cdot s_t^{(k)} \odot m_k)}{2\delta} \cdot s_{t,i}^{(k)} \quad (1)$$

where m_k is the indicator mask for block B_k and $\delta > 0$ is the perturbation magnitude. This requires $2K$ function evaluations per step. The key insight is that by perturbing only $K \ll D$ dimensions simultaneously, the cross-contamination noise from uncorrelated parameters is bounded by the block fraction K/D .

3.2 Temporal Aggregation

The raw gradient estimate g_t is noisy due to cross-block interference. DGE injects it into an Adam-style exponential moving average:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \quad (2)$$

with bias correction $\hat{m}_t = m_t / (1 - \beta_1^t)$ and $\hat{v}_t = v_t / (1 - \beta_2^t)$. Under the assumption of local stationarity, the true gradient $\nabla f(x)$ contributes with consistent sign across steps (accumulating constructively in m_t), while cross-perturbation noise has random sign (canceling destructively). After T steps, the signal grows as $\mathcal{O}(T)$ while noise grows as $\mathcal{O}(\sqrt{T})$, yielding SNR improvement as $\mathcal{O}(\sqrt{T})$.

3.3 Direction-Consistency Learning Rate

To suppress anomalous noise bursts in pathological landscapes, DGE computes a per-parameter confidence coefficient over a window of T steps:

$$C_{t,i} = \left| \frac{1}{T} \sum_{k=0}^{T-1} \text{sgn}(g_{t-k,i}) \right| \in [0, 1] \quad (3)$$

Parameters dominated by noise exhibit random sign walks, driving $C_{t,i} \rightarrow 0$ via the law of large numbers. Parameters with consistent gradient direction maintain $C_{t,i} \rightarrow 1$. The update becomes:

$$x_{t+1} = x_t - \alpha \cdot C_t \odot \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad (4)$$

3.4 Dual Sign-EMA (DS-EMA)

The sliding-window consistency mask (Eq. 3) has two limitations: (1) edge effects where old signs exit the window abruptly, and (2) $\mathcal{O}(T \cdot D)$ memory for storing the sign buffer. We replace it with a MACD-inspired dual exponential mechanism:

$$e_t^{\text{fast}} = (1 - \alpha_f) e_{t-1}^{\text{fast}} + \alpha_f \text{sgn}(g_t) \quad (5)$$

$$e_t^{\text{slow}} = (1 - \alpha_s) e_{t-1}^{\text{slow}} + \alpha_s \text{sgn}(g_t) \quad (6)$$

with $\alpha_f = 0.3$ (fast) and $\alpha_s = 0.05$ (slow). The consistency mask becomes a crossover gate:

$$D_t = \mathbb{K}[\text{sgn}(e_t^{\text{fast}}) = \text{sgn}(e_t^{\text{slow}})] \odot |e_t^{\text{slow}}| \quad (7)$$

When the fast and slow EMAs disagree on the gradient direction, the parameter is likely noise-dominated and its update is suppressed. When they agree, the update is scaled by the slow EMA's magnitude, reflecting long-term confidence. This requires only $\mathcal{O}(2D)$ additional memory—a 90% reduction from the $\mathcal{O}(T \cdot D)$ sliding window.

Algorithm 1 Denoised Gradient Estimation (DGE)

Require: Objective f , initial point $x_0 \in \mathbb{R}^D$, block size K , learning rate α , perturbation δ , total steps T

- 1: Initialize: $m_0 = v_0 = 0$, $e_0^{\text{fast}} = e_0^{\text{slow}} = 0$
- 2: **for** $t = 1, \dots, T$ **do**
- 3: $\pi_t \leftarrow$ random permutation of $\{1, \dots, D\}$
- 4: Partition into K blocks: $B_1, \dots, B_K$ via π_t
- 5: **for** each block B_k **do**
- 6: Sample $s^{(k)} \sim \text{Rademacher}(\{-1, +1\}^{|B_k|})$
- 7: $g^{(k)} \leftarrow \frac{f(x + \delta \cdot s^{(k)} \odot m_k) - f(x - \delta \cdot s^{(k)} \odot m_k)}{2\delta} \cdot s^{(k)}$
- 8: **end for**
- 9: Assemble g_t from all $g^{(k)}$
- 10: Update EMA: $m_t = \beta_1 m_{t-1} + (1 - \beta_1)g_t$, $v_t = \beta_2 v_{t-1} + (1 - \beta_2)g_t^2$
- 11: Compute DS-EMA mask D_t via Eqs. 5–7
- 12: $x_t \leftarrow x_{t-1} - \alpha \cdot D_t \odot \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$
- 13: **end for**
- 14: **return** x_T

3.5 Block Size Selection

The block size K controls the bias-variance tradeoff: smaller K reduces variance per coordinate but requires more steps to cover the full parameter space. Through systematic ablation across five scaling formulas ($\mathcal{O}(1)$, $\mathcal{O}(\log D)$, $\mathcal{O}(\sqrt{D})$, $\mathcal{O}(D/100)$, $\mathcal{O}(D/10)$), we find that $K = \mathcal{O}(\sqrt{D})$ achieves the best tradeoff, reaching 90.49% accuracy on MNIST at 500K evaluations—outperforming both smaller ($\mathcal{O}(\log D)$: 87.14%) and larger ($\mathcal{O}(D/100)$: 89.49%) block sizes.

The complete algorithm is summarized in Algorithm 1.

4 Theoretical Analysis

Theorem 1 (Unbiasedness). *Under first-order Taylor approximation $f(x + \Delta) \approx f(x) + \nabla f(x)^\top \Delta$, the block perturbation estimator (Eq. 1) is an unbiased estimator of the true gradient: $\mathbb{E}[g_{t,i}] = \nabla f(x)_i$ for all active coordinates i where $m_{t,i} = 1$.*

Proof. Expanding the symmetric difference for an active coordinate i :

$$\begin{aligned}
g_{t,i} &= \frac{f(x + \delta \cdot s \odot m) - f(x - \delta \cdot s \odot m)}{2\delta} \cdot s_i \\
&\approx \frac{2\delta \sum_j \nabla f(x)_j s_j m_j}{2\delta} \cdot s_i = \sum_j \nabla f(x)_j s_j s_i m_j \\
&= \nabla f(x)_i s_i^2 + \sum_{j \neq i} \nabla f(x)_j s_j s_i m_j
\end{aligned}$$

Since $s_i \in \{-1, +1\}$, we have $s_i^2 = 1$. For $j \neq i$, s_j and s_i are independent Rademacher variables with $\mathbb{E}[s_j s_i] = 0$. Therefore $\mathbb{E}[g_{t,i}] = \nabla f(x)_i$. $\square$

Lemma 2 (Variance Bound). *The conditional variance of the block perturbation estimator for an active coordinate i is:*

$$\text{Var}(g_{t,i}) = \sum_{j \neq i} (\nabla f(x)_j)^2 m_j \leq \frac{K}{D} \|\nabla f(x)\|^2 \quad (8)$$

Proof. The variance follows from the independence of Rademacher variables: $\text{Var}(g_{t,i}) = \mathbb{E}[(\sum_{j \neq i} \nabla f(x)_j s_j s_i m_j)^2] = \sum_{j \neq i} (\nabla f(x)_j)^2 m_j$. Since exactly K of the D dimensions are active ($\sum m_j = K$), the upper bound follows from $\sum_{j \neq i} (\nabla f(x)_j)^2 m_j \leq \max_j (\nabla f(x)_j)^2 \cdot K \leq (K/D) \|\nabla f\|^2$. $\square$

Table 1: Test accuracy on full MNIST (60K/10K). DGE achieves 96.3% of Adam’s performance without any gradient information.

Method	Type	Accuracy	Std
SPSA	Zeroth-order	20.87%	$\pm 11.83\%$
MeZO	Zeroth-order	20.87%	$\pm 11.83\%$
PureDGE	Zeroth-order	93.00%	$\pm 0.24\%$
DGE (DS-EMA)	Zeroth-order	94.36%	$\pm 0.23\%$
SGD + momentum	Backprop	97.78%	$\pm 0.04\%$
Adam	Backprop	98.00%	$\pm 0.04\%$

This represents a D/K -fold **variance reduction** compared to SPSA, where all D dimensions are perturbed simultaneously. For $D = 100,000$ and $K = 128$, this yields a $781\times$ reduction.

Corollary 3 (SNR Improvement). *Under local stationarity of $\nabla f(x)$, the signal-to-noise ratio of the temporally filtered estimate improves as $\text{SNR}(T) = \mathcal{O}(\sqrt{T})$.*

Proof. The true gradient signal accumulates linearly: $\mathbb{E}[m_t] \propto T \cdot |\nabla f|$. Independent noise accumulates as a random walk: $\text{std}(m_t) \propto \sqrt{T} \cdot \sigma$. Therefore $\text{SNR} = \mathbb{E}[m_t]/\text{std}(m_t) \propto \sqrt{T} \cdot |\nabla f|/\sigma$. $\square$

5 Experiments

We evaluate DGE on differentiable and strictly non-differentiable architectures, comparing against SPSA, MeZO, and backpropagation baselines (SGD, Adam).

5.1 Setup

Architecture. MLP with layers $[784 \rightarrow 128 \rightarrow 64 \rightarrow 10]$ (109,386 parameters) unless otherwise noted.

Dataset. MNIST [LeCun et al., 2015]: full (60K/10K) or 3K-subset (3,000/600) as specified.

Budget. DGE methods receive 3M function evaluations for full MNIST, 800K for the 3K-subset. Back-propagation baselines train for 30 epochs.

Hyperparameters. $K = [1024, 128, 16]$ per layer, $\delta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$. Learning rate: 0.05 (ZO methods), 10^{-3} (Adam), 0.01 (SGD). All results averaged over 3 seeds (full MNIST) or 6 seeds (3K-subset) with 95% confidence intervals.

5.2 Differentiable MNIST

Table 1 presents results on the full MNIST dataset.

SPSA and MeZO collapse to $\sim 21\%$ accuracy (near random guessing for 10 classes), confirming the fundamental dimensionality barrier. PureDGE with block perturbations alone achieves 93.00%, demonstrating that variance reduction via blocking is the primary enabler. The DS-EMA consistency mask provides an additional +1.36pp improvement, reaching 94.36%—only 3.64pp below Adam.

Figure 1 shows the convergence curves. DGE separates from SPSA within the first 30K evaluations and maintains a consistent lead throughout training.

5.3 Non-Differentiable Networks

DGE’s primary advantage emerges in settings where backpropagation cannot operate. Table 2 reports results on architectures with zero analytical gradients, without using straight-through estimators.

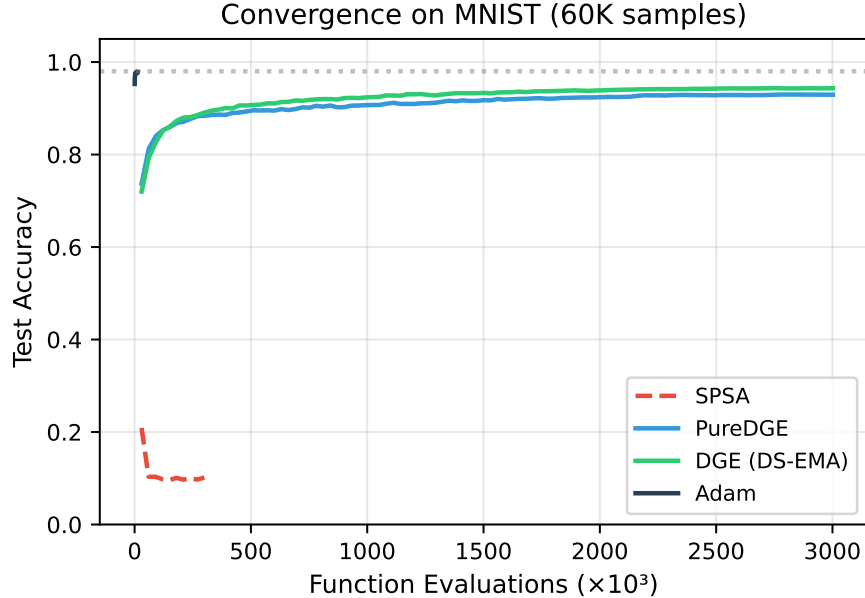


Figure 1: Convergence on full MNIST (60K samples). DGE (DS-EMA) achieves 94.36% while SPSA and MeZO collapse to $\sim 21\%$.

Table 2: Non-differentiable architectures. Adam fails completely without STE; DGE operates natively on discrete landscapes.

Architecture	Adam	DGE
Sign activations (784-32-10)	61.20%	73.20%
INT8 full quantization (256 levels)	8.40%	82.20%
INT4 full quantization (16 levels)	9.30%	77.80%

Sign activations. The step function $\text{sign}(x)$ has zero derivative everywhere. Adam can only train the final linear layer (61.20%), while DGE penetrates through the non-differentiable activations to optimize hidden layers, achieving 73.20%.

INT8 quantization. With 256 discrete levels, the quantized network provides no gradient signal. DGE achieves 82.20%—within 12pp of the continuous baseline—demonstrating native quantization-aware training without STE.

INT4 quantization. At extreme 4-bit precision (16 levels), DGE still reaches 77.80%, establishing its viability for ultra-low-power hardware.

Figure 2 visualizes the dramatic performance gap between DGE and Adam in these regimes.

5.4 Synthetic Benchmarks

Table 3 compares DGE variants on standard optimization test functions (128D, 200K evaluations, 5 seeds).

DS-EMA consistently outperforms the sliding-window approach, particularly on the Rotated Quadratic where the SMA variant degrades by 3 orders of magnitude due to edge effects in the sign buffer.

5.5 Ablation Study

Figure 3 decomposes the contribution of each DGE component on full MNIST.

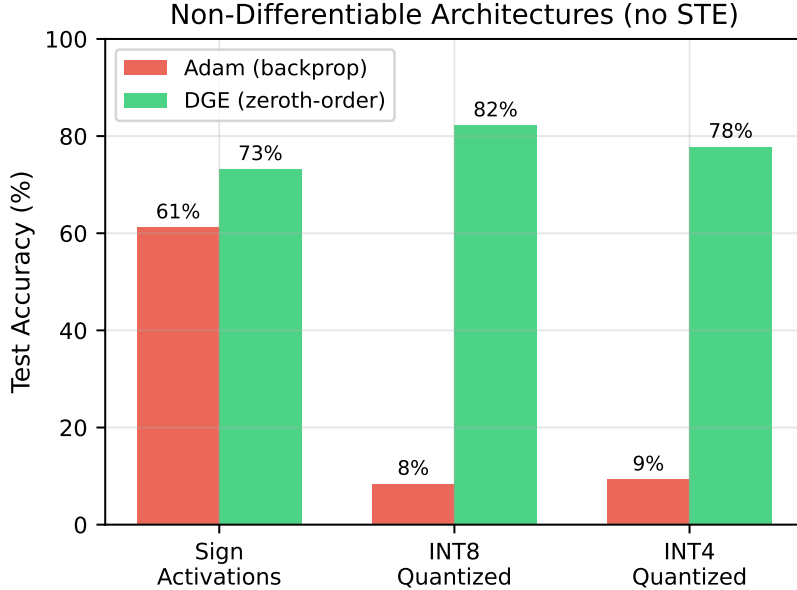


Figure 2: Non-differentiable architectures (no STE). DGE substantially outperforms Adam across all discrete settings.

Table 3: Final loss on synthetic benchmarks (lower is better). DS-EMA achieves best or near-best on all functions.

Function	PureDGE	Consist. (T=20)	DS-EMA
Rosenbrock	1.40×10^2	1.14×10^2	1.06×10^2
Rotated Quadratic	2.82×10^{-12}	3.48×10^{-9}	1.49×10^{-12}
Ellipsoid ($\kappa=10^6$)	7.51×10^3	7.43×10^2	2.99×10^2
Sphere (8192D)	6.81×10^{-39}	5.85×10^{-29}	2.43×10^{-39}

The block perturbation mechanism is the primary enabler (+72.13pp over SPSA). Temporal EMA provides the second largest gain (+5pp), while the DS-EMA consistency mask adds the final +6.36pp. The three mechanisms are complementary: removing any one causes significant degradation.

5.6 Convergence Analysis (6 Seeds)

Figure 4 shows convergence on the MNIST-3K subset with 6 independent seeds, providing statistical rigor.

The 95% confidence intervals do not overlap: PureDGE [80.4%, 84.1%] vs. DGE+Consistency [86.6%, 88.6%], confirming the improvement is statistically significant. The consistency mask also reduces inter-seed variance by 45% (std: 2.25% $\rightarrow$ 1.23%).

6 Discussion

Limitations. DGE has two identified failure modes. First, *deep discrete networks* suffer from the butterfly effect: a single bit-flip in early layers can invert the state of all subsequent layers, destroying the temporal stationarity required by the EMA filter. This caps accuracy at $\sim 75\%$ for deep sign-activation networks. Second, in *fully differentiable settings*, DGE does not compete with backpropagation in computational efficiency—Adam requires $\mathcal{O}(1)$ gradient evaluations per step via the backward pass, while DGE requires $\mathcal{O}(K)$ function evaluations.

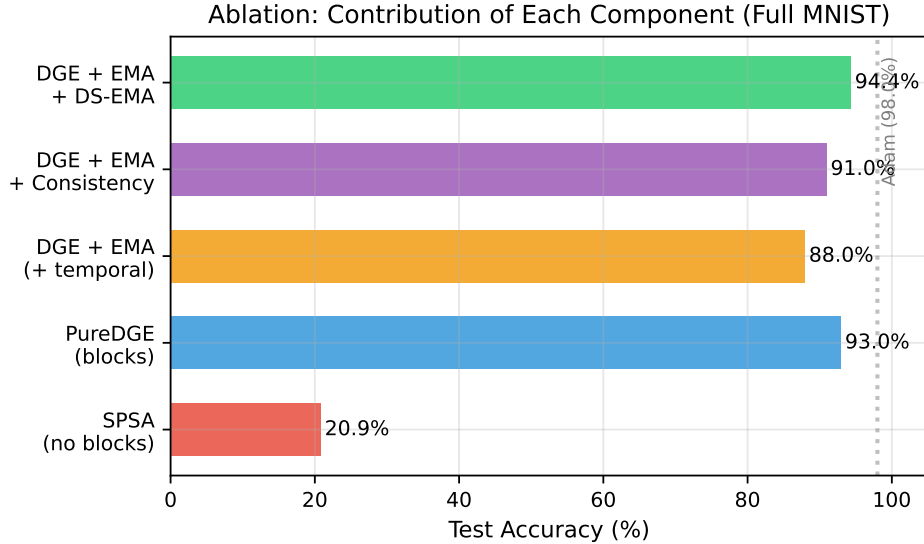


Figure 3: Component ablation on full MNIST. Each mechanism contributes multiplicatively to the final accuracy.

Practical recommendations. Based on our empirical analysis, we recommend: (1) use $K = \mathcal{O}(\sqrt{D})$ block sizes for neural network architectures; (2) set learning rate $\alpha \leq 0.1$ for DS-EMA consistency to be effective (higher rates cause sign oscillations that the mask interprets as noise); (3) use cosine annealing for both δ and α to transition from exploration to exploitation.

Applications. DGE is particularly well-suited for: (i) neuromorphic hardware training where backpropagation is physically impossible [Schuman et al., 2022]; (ii) quantization-aware training of INT4/INT8 networks without STE heuristics [Banner et al., 2019]; (iii) black-box optimization of API-based systems or physics simulators; and (iv) fine-tuning of LLMs in memory-constrained settings, complementing MeZO with block-level variance reduction.

7 Conclusion

We presented Denoised Gradient Estimation (DGE), a zeroth-order optimization algorithm that trains neural networks using only forward passes. Through block perturbations, temporal EMA filtering, and the novel Dual Sign-EMA consistency mask, DGE overcomes the dimensionality curse that limits traditional zeroth-order methods.

DGE achieves 94.36% on MNIST without gradients (96.3% of Adam), and dominates non-differentiable settings where backpropagation fails entirely (82% on INT8 quantized networks vs. 8% for Adam). The algorithm requires no assumptions about differentiability, making it a universal optimizer for discrete, quantized, and black-box systems.

Future work includes scaling DGE to larger architectures via GPU-batched perturbation evaluation, integrating block-adaptive perturbation magnitudes for hybrid discrete-continuous networks, and exploring DGE as a complement to backpropagation for regularizing non-stationary training dynamics.

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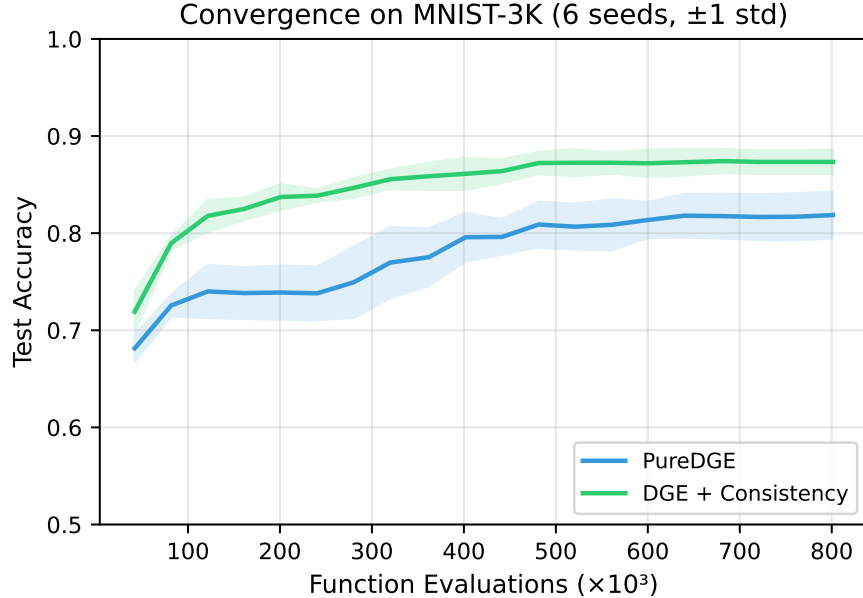


Figure 4: Convergence on MNIST-3K (6 seeds, mean $\pm$ 1 std). DGE+Consistency achieves $87.58\% \pm 1.23\%$ vs. PureDGE $82.22\% \pm 2.25\%$.

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